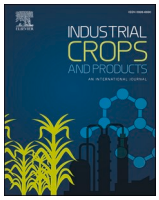
Industrial Crops & Products 236 (2025) 122042



Contents lists available at ScienceDirect

Industrial Crops & Products

journal homepage: www.elsevier.com/locate/indcrop





Pretreatment with alkyl trimethylammonium bromide-based deep eutectic solvent cetyltrimethyl ammonium bromide/lactic acid/Al^{3+} benign system for the efficient production of furfural

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ARTICLE INFO

Keywords:
Deep eutectic solvent CTAB/LAC/Al^{3+}
Furfural
Structure characterization
Molecular dynamics simulation
Pretreatment
Valorization

ABSTRACT

Furfural, which can be derived from renewable biomass, is a natural precursor for producing of drugs, biofuels, biochemicals, and polymers. This research synthesized a ternary deep eutectic solvent (DES) for the efficient conversion of rapeseed straw into furfural. The DES was firstly synthesized by mixing surfactant cetyltrimethylammonium bromide (CTAB), lactic acid (LAC) and metal chloride ($AlCl_3$), which exhibited 29.3 mN/m of surface tension and 1293.8 cp of viscosity. The CTAB/LAC/Al^{3+}-water reaction system was used to transform rapeseed straw for producing furfural through a two-step reaction method, with a furfural yield reaching 71.6 %. The structure characterization alteration of rapeseed straw before and after CTAB/LAC/Al^{3+} action was explored by using SEM, CLSM, BET, FT-IR and other methods. The interaction between the biomass model and solvents in the CTAB/LAC/Al^{3+}-water reaction system was revealed through molecular dynamics simulation. By combining the CTAB/LAC/Al^{3+} properties, chemical component changes of biomass, biomass surface and structure alterations, and molecular dynamics analysis, the catalysis mechanism for treatment and transformation of rapeseed straw into furfural was interpreted. This work involving the application of ternary DES CTAB/LAC/Al^{3+}-water benign system to transformation of waste biomass provided an efficient way for production of furfural.

1. Introduction

The biomass refining plant for lignocellulose transforms agricultural waste into high value-added products and is key to achieving a green transition (Sriariyanun et al., 2023). Furfural (FAL) derived from lignocellulosic biomass containing pentosans has been identified as one of the top ten most valuable biobased chemicals (Li et al., 2021a), which is an important platform molecule for manufacturing functional derivatives, drug intermediates, solvents, additives, polymers, plastics, fuels, etc. (Morais et al., 2021). The biorefining technology for producing furfural has significant advantages over traditional single production lines, mainly reflected in its comprehensiveness, efficiency, and sustainability (Areeya et al., 2024). FAL production faces issues such as high cost of raw materials, inefficient reactions, and potential environmental pollution (Amesho et al., 2023). There is an urgent need for efficient and benign processes to acquire high productivity of FAL to meet market demand. Homogeneous and heterogeneous solid chemocatalysts (such as acids and metal salts) are extensively utilized to produce FAL in water, organic solvent, ionic liquid (Ahn et al., 2024), ionic liquids, etc (Cousin et al., 2022; Hua et al., 2021).

Deep eutectic solvents (DESs) are considered to green reaction medium and chemocatalysts for FAL production due to their biocompatibility, non-volatility, good thermostability and high catalytic activity (Wang et al., 2023a). In addition, from the perspective of green manufacturing, the preparation of DES is simple (Baraka et al., 2025), and the production cost is low (Falahudin and Insin, 2024). ChCl/1, 2-Propylene glycol/4-Methyl-2-pentanone treated eucalyptus to acquire the highest FAL yield of 65.0 % and saccharification efficacy of 92.2 % (Yue et al., 2024). Corn cobs were catalyzed in ChCl:MA-H_2O (40 wt%:60 wt%, 170 °C) in 20 min to obtain FAL yield in 63.0 % (Di et al., 2023). After the pretreatment of corn straw with a DES composed of ammonium chloride and monoethanolamine, the yields of glucose and xylose reached 96 % after 24 h of enzymatic hydrolysis (Wang et al., 2025). Surfactants are substances harboring hydrophobic

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https://doi.org/10.1016/j.indcrop.2025.122042
Received 19 August 2025; Received in revised form 29 September 2025; Accepted 30 September 2025
Available online 7 October 2025
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hydrocarbon-based "tail" regions and hydrophilic "head" groups, which have specified functions in biomass valorization (Bathke et al., 2025). Surfactants are able to effectively interact with the hydrophobic groups present in lignin molecules, consequently promoting the solubility of lignin in liquid media and effectively hampering its re-deposition to the surface of the biomass (Xu et al., 2023). The results of the surfactant-assisted acidic glycerol pretreatment of sugarcane bagasse indicate that in the presence of surfactants, the consumption of acid is reduced, and the surfactants induce structural modifications of the biomass (Song et al., 2025). Surfactant (Tween 80) modified the residual lignin through etherification, reducing phenolic hydroxyl groups, alleviating the non-productive adsorption between lignin and enzymes, and improving the hydrolytic properties of the substrate (Song et al., 2024). N-Alkyl trimethylammonium bromide (C_nTAB) as a cationic surfactant with good safety and controllable cost, which can be used for preparing Fe^{3+}-based DES for the biomass transformation and the production of FAL (3.6 g/L, 173 °C, 82 min) (Tang et al., 2024). In terms of cost, C_nTAB is higher than conventional HBDs like choline chloride. C_nTAB has an advantage in biomass pretreatment due to the nature of its surfactant, as it prevents the re-deposition of pseudo-lignin on the biomass surface (Tang et al., 2024). Acidic DESs can remarkably cleave ether bonds, considerably depolymerize lignin, apparently hydrolyze hemicellulose and promote the formation of FAL (e.g., CTAC:LAC: 3.8 g/L, 160 °C, 60 min; TTAB/LAC/Fe^{3+}: FAL 3.8 g/L, 163 °C, 62 min) (Li and Wu, 2025). Conventional FAL production is mainly obtained through the chemocatalysis with inorganic acids (sulfuric acid, hydrochloric acid, etc.), which generates a large amount of solid and liquid waste, and the FAL yield is not ideal (Liu et al., 2025). Difference from inorganic acids, organic acids manifest significant advantages because they are less corrosive (Farooq et al., 2020). Binary DES ChCl:*p*-Toluene sulfonic acid was used to pretreat biomass (120 °C, pH 1.0, 90 min) to produce FAL in 85 % yield (Arora et al., 2021). In addition, studies have manifested that supplementing metal chlorides ($AlCl_3$, $FeCl_3$, $CuCl_2$, etc.) into the acidic DES system can considerably augment the generation of FAL yield. Metal chlorides can elevate their catalytic activity by efficiently pairing with electrons, providing a richer Cl^- and enhancing the degradation of hemicellulose in the action of acidic DES, consequently promoting FAL generation (Ji et al., 2025). This is due to the interaction of positively and negatively charged ions in metal chlorides with the reactants (Rostami et al., 2025). Accordingly, metal chloride is added as the third component of the organic acid-based DES to promote the production of FAL.

Rapeseed straw (RS) is a widely available waste stream, with RS production increasing from 36 to 72 million tonnes between 2001 and 2020 (Abbasi-Riyakhuni et al., 2025). RS is primarily composed of lignin, hemicellulose, cellulose, and other components. The stubbornness of RS structure limits its conversion into high-value chemicals and fuels (Raj and Singh, 2025), and the complexity of its composition makes it difficult to effectively separate and utilize for high-end applications (Wang et al., 2022). Therefore, efficient and tailored systems are needed to process challenging biomass such as RS. Ternary DES involves three components, which provide greater flexibility in designing solvents with specific properties such as polarity, viscosity, and thermal stability. Ternary DES commonly uses ChCl as the HBA, polyvalent alcohols as the HBD, and toluene sulfonic acid (Chen et al., 2019), oxalic acid, and lactic acid as acidic HBDs (Narayanan et al., 2024). Carboxylic acid-based DES is more effective than polyvalent alcohol-based DES in biomass pretreatment (Chen et al., 2019). To realize the green and sustainable utilization of RS, it can attempt to develop surfactant-based acidic ternary DESs that combines C_nTAB, organic acids and metal chlorides to effectually catalyze waste RS into valuable FAL in this work. C_nTAB is an HBA/surfactant, organic acids are HBD/Brønsted acids, and metal chlorides are Lewis acid catalysts. A variety of acidic C_nTAB-based DESs were firstly prepared, and their surface tension, viscosity, and K-T parameters were measured. The optimal C_nTAB-based DES was selected according to the FAL yield, and the process parameters such as ternary DES composition ratio, ternary DES loading, reaction temperature, and reaction duration were optimized to augment FAL synthetic efficacy. Eventually, the valorization mechanism of RS into FAL was deeply analyzed by using scanning electron microscopy (SEM), Laser confocal microscopy (CLSM), Fourier transform infrared spectroscopy (FT-IR) and other methods of characterization and molecular dynamics simulation. The efficacy and eco-friendliness of this FAL production process from waste biomass was well clarified.

2. Materials and methods

2.1. DES preparation

Different C_nTABs [dodecyl trimethyl ammonium bromide (DTAB), tetradecyl trimethyl ammonium bromide (TTAB), cetyltrimethyl ammonium bromide (CTAB), octadecyl trimethyl ammonium bromide (OTAB)], organic acids [lactic acid (LAC), formic acid (FA), acetic acid (HAC), propionic acid (PA), butyric acid (BA) and levulinic acid (LeA)] and metal chloride salts [$AlCl_3$, $FeCl_3$, $FeCl_2$, $CuCl_2$, $ZnCl_2$, $MnCl_2$, $CrCl_3$, $SnCl_4$] were mixed in the designed molar ratio and stirred in an oil bath pot at 90 ℃ to achieve the preparation of a variety of C_nTAB-based acidic ternary DESs.

2.2. Synthesis and determination of FAL

One-step synthesis of FAL: 3.0 g RS was immersed in 40 mL of CTAB/LAC/Al^{3+}-water (water content of 0–90 wt%) in a reactor (140–190 ℃) in 5–50 min. Afterwards, the reaction solution was filtered, and the FAL concentration was determined.

Two-step synthesis of FAL: 3.0 g RS was soaked in 40 mL of CTAB/LAC/Al^{3+}-water (50 wt%) in a reactor (140–190 ℃) for reacting in 5–50 min to obtain a biomass-derived xylose solution. Afterwards, the reaction solution was filtered, and the separated xylose solution was transformed to FAL in another reactor (170–190 ℃) for 0.5–40 min. Eventually, the FAL and RS residues were determined. The FAL yield calculation is based on the theoretical pentose yield of hemicellulose in RS.

2.3. Statistical analysis

In this work, each experiment was conducted three times. Meanwhile, the experimental results are presented as mean ± standard deviation of the three independent replicates. ANOVA and Tukey tests were performed using SPSS (IBM, version 25).

3. Results and discussion

3.1. Screening of C_nTAB-based ternary DES

To comprehensively select the most suitable acidic CnTAB-based DES for RS, the three components of DES were screened by a single-factor method (170 ℃, 60 min), including C_nTAB, organic acids and metal chlorides. Four types of C_nTAB with different alkyl chain lengths (12−18) were synthesized with lactic acid (LAC) for preparing binary DESs, and the effects of C_nTAB types on FAL yield were estimated (**Fig. 1** a). The yield of biomass-derived FAL continued to elevate as the C_nTAB strand length increases from 12 to 18. Research has shown that as the C_nTAB carbon chain decreases from 18 to 12, the xylan removal rate increases from 65.5 % to 73.4 % (Tang et al., 2025). At lower carbon chain lengths, the xylan removal rate increases. As shown in **Table 1**, as the C_nTAB carbon chain decreases from 18 to 12, the surface tension decreases from 30.2 mN/m to 28.1 mN/m. Relatively low surface tension helps the hydrogen ions diffuse in the acidic DES and penetrate the lignin and hemicellulose of RS (Chen et al., 2024). As the carbon chain decreases, the pretreatment intensity increases, and the xylan removal rate increases, resulting in a reduction in the amount of raw materials

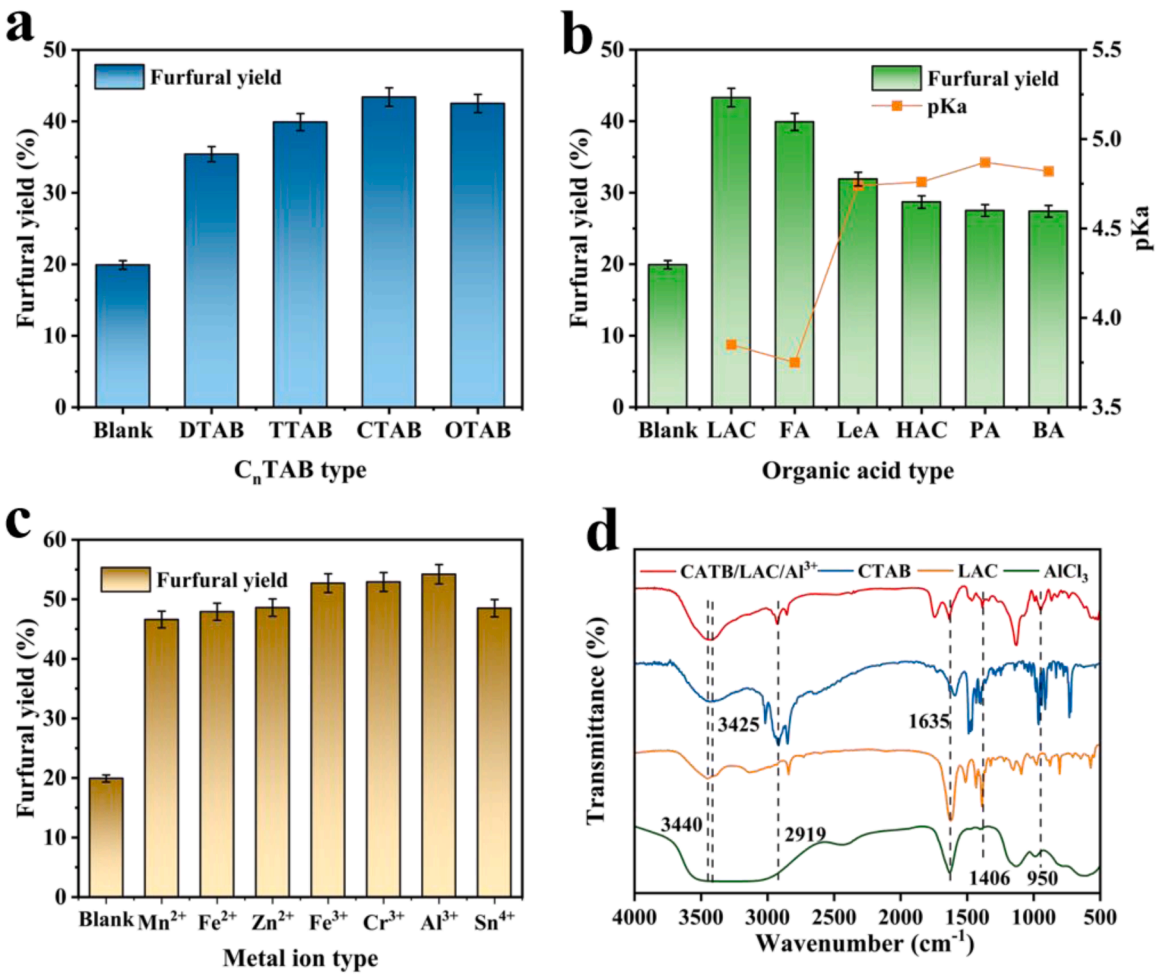


Fig. 1. Effect of C_nTAB type on FAL yield [170 ℃, 60 min, C_nTAB/LAC 1:4 (mol/mol)] (a); Effects of organic acid types on FAL yield [170 ℃, 60 min, CTAB/organic acid 1:4 (mol/mol)] (b); Effects of metal ion types on furfural yield [170 ℃, 60 min, CTAB/LAC/metal ion 1:4:0.04 (mol/mol/mol)] (c); The FT-IR image of CTAB/LAC/Al^{3+} components and CTAB/LAC/Al^{3+} (d).

Table 1
The physicochemical properties of the C_nTAB-based DES.

No.	DES type	Viscosity, cp	Surface tension, mN/m	α	β	π*
1	DTAB/LAC	350.3 ± 12.3	28.1 ± 0.8	1.90 ± 0.06	1.65 ± 0.05	0.01 ± 0.003
2	TTAB/LAC	848.2 ± 30.9	28.8 ± 0.7	1.76 ± 0.05	1.62 ± 0.05	0.04 ± 0.001
3	CTAB/LAC	1632.5 ± 32.6	29.4 ± 0.8	1.07 ± 0.03	2.50 ± 0.07	0.06 ± 0.001
4	OTAB/LAC	1983.5 ± 40.7	30.2 ± 0.8	1.02 ± 0.03	1.39 ± 0.03	0.11 ± 0.003
5	CTAB/LAC/Al^{3+}	1293.8 ± 28.7	29.3 ± 0.7	0.49 ± 0.01	3.13 ± 0.09	1.20 ± 0.04

available for FAL production, which leads to a decrease in FAL yield. CTAB-based DES was used for action of RS, and the FAL yield reached 43.3 %. Meanwhile, the FAL yield of OTAB-based DES reached 42.5 %. However, OTAB-based DES is in a solid state at room temperature, which might increase the difficulty of transfer and separation before and after action (Tang et al., 2025). Consequently, CTAB was selected as the hydrogen-bond-donor for the action of RS.

Afterwards, six different organic acids (FA, HAC, PA, BA, LAC, and LeA) commonly used in catalyzing RS were mixed with CTAB. The CTAB-based DES of different organic acids were screened to obtain the most suitable binary C_nTAB-based DES for RS action. As indicated in Fig. 1 b, LAC-CTAB had the best catalytic effect, with a FAL yield of 43.3 %. While BA-CTAB had the lowest FAL yield (27.4 %), probably due to its relatively high pKa value that disrupted the lignocellulose structure causing more severe damage. The loading of H^+ ions can impact the formation of FAL by influencing the hydrolysis of xylans (Yang et al., 2022). As evidenced in Fig. 1 b, the FAL yield decreases with the increase of pKa, but is not completely inhibited.

Fig. 1 c compares the FAL yields of nine DESs containing different metal chlorides. Studies have showcased that ternary DES containing metal chlorides can obtain higher FAL yields compared with CTAB/LAC. This might be due to the synergistic effect of metal chloride and DES, which can create an acidic environment to augment lignocellulose hydrolysis and further catalyzes the synthesis of FAL (Huang et al., 2022; Tang et al., 2024). CTAB/LAC/Al^{3+} action yielded FAL in the highest yield of 54.2 %. The increase in FAL yield of low-valence metal chlorides was generally lower than that of high-valence metal chlorides, revealing that high-valence metal chlorides might render a stronger electric field gradient and consequently augmenting interaction with reaction substrates to promote FAL generation (Lee et al., 2022). Fig. 1 c shows the promoting effect of metal chlorides on the generation of FAL as follows: $AlCl_3$ > $CrCl_3$ > $FeCl_3$ > $SnCl_4$ > $ZnCl_2$ > $FeCl_2$ > $MnCl_2$. The Lewis

acid strength is as follows: $AlCl_3$ > $CrCl_3$ > $FeCl_3$> $SnCl_4$ > $ZnCl_2$ > $FeCl_2$ > $MnCl_2$. The ionic radii (pm) are as follows: Al^{3+} (53.0) < Cr^{3+} (61.5) < Fe^{3+} (64.5) < Sn^{4+} (69.0) < Zn^{2+} (74.0) < Fe^{2+} (78) < Mn^{2+} (83.0). It is reported that the higher the ionization energy, the stronger the Lewis acid strength, and the higher the dehydration rate of xylose, leading to an increased yield of FAL (Lee et al., 2022). This study indicates that the promoting effect of metal chlorides on FAL is directly proportional to the strength of the metal chlorides and inversely proportional to the radius of the metal ions. In DES CTAB/LAC/Al^{3+}, Al^{3+} might disrupt the lignocellulose structure and efficiently stripping the hemicellulose component (Huang et al., 2022), and the glycosidic bonds by bonding with glycoside oxygen atoms can be weakened (Cousin et al., 2022), resulting in the enhanced hydrolysis of xylan to xylose and consequently promoting the dehydration of xylose to FAL in the presence of acidic CTAB/LAC/Al^{3+}.

As evidenced in Table S1 (see Supplementary files), the P values for C_nTAB type, Organic acid type, and Metal ion type are all less than 0.001, indicating that the differences between the various levels are statistically significant. After conducting the Tukey test, the yield of FAL under CTAB, LAC, and $AlCl_3$ conditions is significantly higher than under other conditions.

3.2. Characterization of ternary CTAB/LAC/Al^{3+}

FTIR images of the three components of a novel ternary DES, CTAB/LAC/Al^{3+}, were acquired. As evidenced in Fig. 1d, a peak caused by -OH tensile vibration was observed near 3425 cm^{-1}, potentially indicating the formation of a new hydrogen bond network in DES (Xiong et al., 2021). In the DES formed by CTAB/LAC/Al^{3+}, the Br- from CTAB serves as the core acceptor of hydrogen bonds, LAC is the core donor of hydrogen bonds, and $AlCl_3$ greatly enhances the strength and complexity of the hydrogen bond donors in the system through its Lewis acidity and hydrolysis effects. DES has a response peak around 1635 cm^{-1}, which was related to COO^- asymmetric telescopic vibration in LAC (Velmurugan et al., 2022). The peak about 950 cm^{-1} belongs to the C-N group of CTAB (Zouboulis et al., 2008). DES is composed of a mixture of two or more components in a certain stoichiometric ratio. The formation of DES is due to the interaction of hydrogen bonds, without the formation of new chemical bonds (Xiao et al., 2024). After the new DES is formed, no new characteristic absorption peaks will appear in the infrared spectrum, and the absorption peaks corresponding to the original functional groups will not disappear but will change in position, intensity, and shape. There are no peaks disappearing in the infrared spectrum of CTAB/LAC/Al^{3+}, and no new peaks appearing, demonstrating the synthesis of this novel ternary DES.

DES has a high viscosity at room temperature due to its abundant hydrogen bonds (Baraka et al., 2025). The high viscosity of DES hinders fluid flow, impeding the mass transfer of lignocellulosic biomass (Ferreira et al., 2021). The increased energy consumption for pumping due to high viscosity also limits its industrial efficiency and feasibility (Yin et al., 2025). The low viscosity may facilitate the mixing of DES with biomass, thereby facilitating the production of FAL (Zhang et al., 2022). As displayed in Table 1, the viscosity of CTAB/LAC was 1632.5 cp, and the viscosity of CTAB/LAC/Al^{3+} was 1293.8 cp. This decrease in viscosity potentially suggested that CTAB/LAC/Al^{3+}, which has better fluidity, may have a stronger catalytic effect during the reaction process.

During the reaction process, surface tension is also an important parameter for solvents, and lower surface tension is conducive to solvent penetration and mass transfer (Wang et al., 2020). As evidenced in Table 1, the surface tension of CTAB/LAC is 29.4 mN/m, and the surface tension of CTAB/LAC/Al^{3+} is about 29 mN/m, which is considerably lower than that of water (72 mN/m). This indicated that the increased reaction efficacy may also be related to the lower surface tension of CTAB/LAC/Al^{3+}. Compared with the melting point of CTAB/LAC of about 38 ℃, ternary DES containing $AlCl_3$ has a significantly lower freezing point and can remain liquid at room temperature, broadening its potential application prospects.

Kamlet-Taft (K-T) parameters are often used for evaluating DESs, namely α, β, and π*, quantifying the hydrogen bond donor capacity (α), hydrogen bond acceptor capacity (β), and polarization capacity (π*) of DES (Mishra et al., 2023; Zhang et al., 2023). Nile Red, p-nitroaniline, N, N-diethyl-4-nitroaniline, and different DES were mixed to determine their absorption spectra, and the Kamlet-Taft parameters of the solvents were calculated based on the obtained spectral data (Tang et al., 2024). The DESs synthesized by C_nTAB with different chain lengths showed a certain trend of K-T parameters. As revealed in Table 1, upon elevating alkyl chain length from 12 to 18, the α value of DES diminished from 1.90 to 1.02, and the β value raised from 1.65 to 2.50, indicating that the decreased acidity can cause the augmented ability to accept hydrogen-bonds (Baraka et al., 2025). C_nTAB-based binary DESs have low polarity, with π* values of less than 0.11, which may be due to the use of low-polarity C_nTAB as a hydrogen-bond-acceptor in DES synthesis (Baraka et al., 2025). Compared with binary DES without metal chloride, the acidity and alkalinity of ternary bonds elevated. The α and β values of CTAB/LAC/Al^{3+} were different from those of binary DESs without metal chloride, revealing that the DES released H^+ more easily and the solvent's hydrogen-bond receptivity was enhanced. In this study, the alterations in viscosity, surface tension, and Kamlet-Taft (K-T) parameters of CTAB/LAC/Al^{3+} compared to CTAB/LAC were considered, and it is likely that the presence of Al^{3+} in the solvent altered the hydrogen-bond network of CTAB/LAC and avoided entanglement caused by intermolecular interactions. This resulted in a reduction in solvent viscosity and surface tension, requiring higher reaction performance (Huang et al., 2022; Tang et al., 2024).

3.3. Optimization of CTAB/LAC/Al^{3+} action

As evidenced in Table S2 (see Supplementary files), the contents of hemicellulose, lignin and cellulose in RS were 15.1 %, 18.9 % and 30.2 %, respectively. After CTAB/LAC/Al^{3+} action, the content of hemicellulose and lignin decreased to 5.5 % and 13.9 %, respectively, and the cellulose content increased to 50.8 %. 85.7 % of the hemicellulose in RS was converted during the reaction process, and the vast majority of cellulose (71.2 %) was retained in the DES-treated RS residue, manifesting that the CTAB/LAC/Al^{3+} could effectively pretreat biomass, which was likely due to the breaking of the barrier formed by lignin and other factors during its reaction, destroying the stubborn structure of lignocellulose (He et al., 2023).

To further explore the effect of the content of various components in CTAB/LAC/Al^{3+} on the yield of FAL, the effect of the molar ratio (1:3–1:6, mol: mol) of CTAB/LAC in DES was first studied. As illustrated in Fig. 2a, upon undergoing conversion by DES in the molar ratio of CTAB/LAC from 1:3–1:4, the FAL yield elevated from 45.8 % to 54.2 %, which may be due to the fact that increasing LAC ratio can render a higher concentration of protons to destroy the stubborn lignocellulose structure and effectively transform hemicellulose into FAL. Further increasing the ratio from 1:4–1:6, the FAL yield declined, which may be due to the increase in the acidity of the solvent when the LAC ratio increases, resulting in a harsher reaction environment, which not only accelerates the synthesis of FAL, but also accelerates the speed of side-reactions, leading to the generation of humin and other by-products and greatly hampers the increase of FAL yield (Zhao et al., 2021).

Due to the negative effects on human health, the detection and extraction of Al^{3+} are significant from both environmental and clinical perspectives. DES-based pretreatment requires the extraction of metal chlorides from wastewater and solid residues to address the leaching issues of metal chlorides (Mandal and Lahiri, 2022). Cloud Point Extraction (CPE) technology and liquid-liquid extraction are both widely used separation techniques for metal extraction (Mandal and Lahiri, 2022). CPE technology utilizes surfactants to extract metal ions efficiently even at trace levels. The CTAB component in the DES used in this study is also a commonly used surfactant in CPE, which may reduce

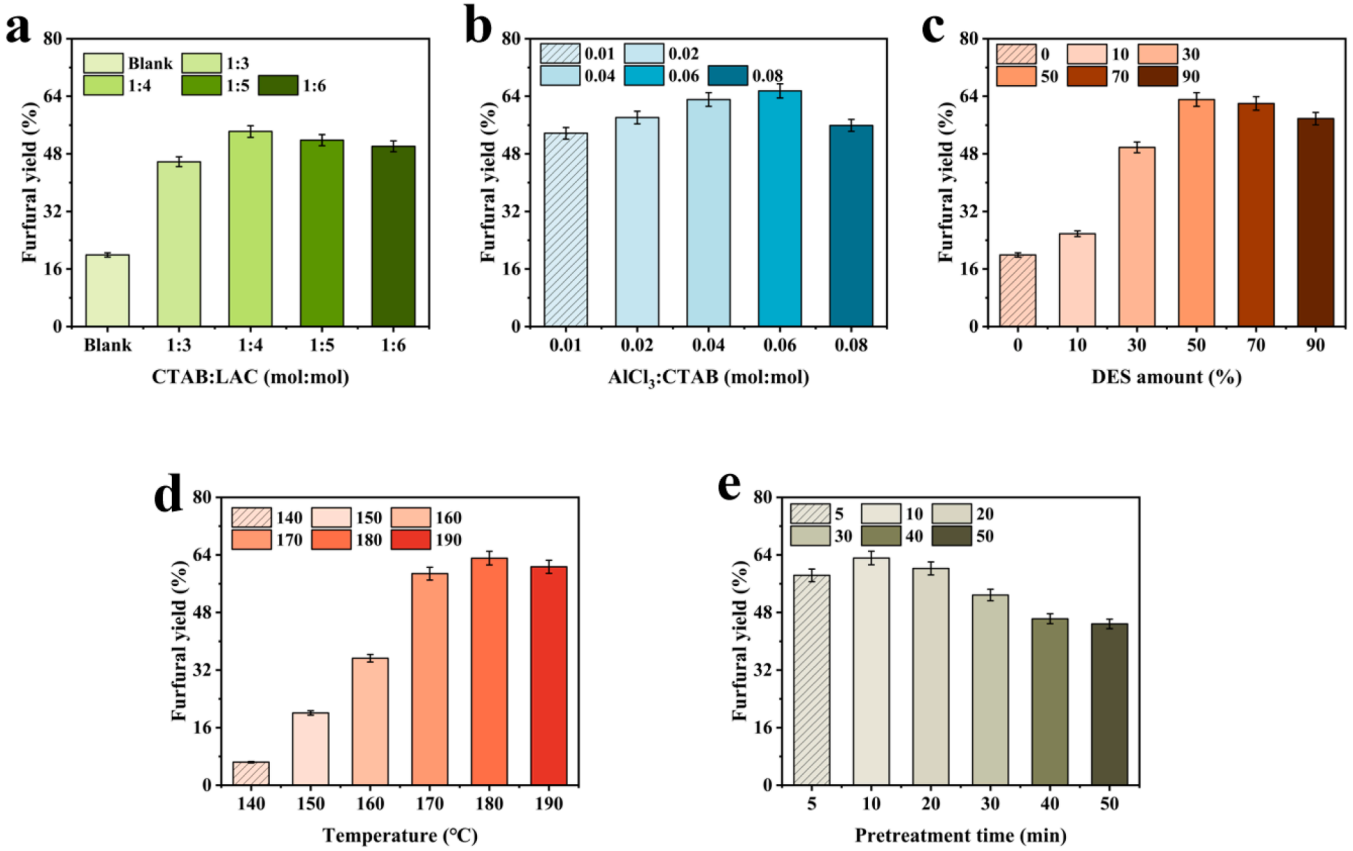


Fig. 2. Effect of CTAB components with different CTAB-to-LAC molar ratio on the FAL generation [170 ℃, 10 min] (a); Effect of $AlCl_3$-to-CTAB molar ratio on the FAL generation [CTAB/LAC 1:4, 170 ℃, 10 min] (b); Effect of DES loading on the FAL generation [CTAB/LAC/$AlCl_3$ 1:4:0.04, 170 ℃, 10 min] (c); Effect of reaction temperature on the FAL generation [CTAB/LAC/$AlCl_3$ 1:4:0.04, DES 50.0 wt%, 10 min] (d); Effect of reaction duration on the FAL generation [CTAB/LAC/$AlCl_3$ 1:4:0.04, DES 50.0 wt%, 180 ℃] (e).

the cost of Al^{3+} extraction.

In this study, the content of $AlCl_3$ in DES was optimized by altering the molar ratio of CTAB to $AlCl_3$ (1:0.01–1:0.08, mol/mol). As evidenced in Fig. 2b, increasing the molar ratio from 1:0.01–1:0.04, the FAL yield elevated from 53.7 % to 63.1 %. For this ratio of 1:0.06, the increase in FAL yield was limited and basically unchanged. Upon reaction with CTAB:$AlCl_3$ in this ratio of 1:0.08, the FAL yield declined to 55.9 %. As the concentration of $AlCl_3$ increased from 2.5 wt% to 5.0 wt%, the yield of FAL increased from 51.9 % to 60.2 % (Lee et al., 2022). This indicates that at low concentrations, the increase in $AlCl_3$ concentration helps it act as a Lewis acid and improves the production efficiency of furfural. This might be due to the presence of Lewis acid in the system, which can catalyze the dehydration and isomerization of xylose, accelerating the synthesis of FAL. As $AlCl_3$ was excessive, the side-reactions intensified, accumulating by-products and reducing FAL yield (Abouelela et al., 2023). In the occurrence of $FeCl_3$ (> 2.8 mM), the FAL concentration dropped considerably due to the acceleration of the side-reaction caused by the high concentration of Lewis acid (Tang et al., 2024). In a concise summary, the CTAB/LAC/Al^{3+} (1:4:0.04) was used for furfural production.

As evidenced in Fig. 2c, the impact of different DES contents (10–90 wt%) on FAL formation were estimated. As the mass fraction of DES raised from 10.0% to 50.0 wt%, the FAL yield also elevated with increasing DES loading, and the FAL yield reached the highest 63.1 % at the mass fraction of 50.0 wt%. Below 50 wt%, the FAL yield diminished considerably. In the acidic reaction systems, excessive acidity will promote the condensation of aldehyde alcohols of furan group compounds formed during the reaction, and other side-reactions such as polycondensation occur, which can weaken the increase of FAL concentration (Zhou et al., 2025). The hydrolysis of hemicellulose requires the participation of water molecules, and the appropriate amount of water molecules promotes the synthesis of FAL, and excess water will reduce FAL generation (Syakir Bahazin et al., 2025). Therefore, the optimal CTAB/LAC/Al^{3+} dose was 50.0 wt%.

In the catalytic synthesis of FAL from lignocellulose, reaction duration and temperature are widely considered to be two the most critical process parameters affecting FAL yield (Lin et al., 2021). Thus, the effects of temperature (140–190 ℃) and duration (5–50 min) on FAL yield were estimated on the FAL generation. As the reaction temperature increased from 140 ℃ to 180 ℃, the FAL yield increased significantly from 6.5 % to 63.1 % (Fig. 2d). Under 190 ℃, the formation rate of FAL is potentially lower than the side reaction rate, resulting in a decrease in FAL yield. This may be because too high temperature disrupting the ordered hydrogen bond network structure of DES itself, making it unable to function effectively, which leads to a decrease in FAL production efficiency and an increase in competitive side reactions (Cousin et al., 2022). The derivatives yield of cellulose and hemicellulose detected in the preprocessed reaction solution is relatively low (formic acid 0.1–0.2 g/L, acetic acid 0.6–1.1 g/L). In the ChCl:Gly-water (DES 7.5 vol%), biomass-derived *D*-xylose was used as substrate to form FAL at 180 ℃, with a FAL yield of 58.3 % (Li et al., 2021b). As evidenced in Fig. 2e, the effect of reaction time had a profound impact on FAL yield. Extending reaction duration from 5 to 10 min, the FAL yield elevated from 56.3 % to 63.1 %. Over 10 min, the yield of FAL continued to decline, which was likely to be closely related to the formation of by-products and the degradation of FAL.

The production process of FAL can be conducted via one-step method and two-step method. The two-step method is composed of

hemicellulose hydrolysis and pentose dehydration/cyclization, which has the advantages of obtaining high FAL yield and diminishing side-reactions (Bao et al., 2024). DES (choline chloride-glycol-proton acid) achieved a FAL yield of 36.4 % in one step (Zhai et al., 2025). Through the catalysis with 0.5 wt% H_2SO_4 at 170 °C, the yield of FAL produced in a one-step was 69 % (Ntimbani et al., 2022). However, the disadvantage of one-step production of FAL is that FAL yield is low or corrosive inorganic acid is used (Zhou et al., 2025). Consequently, a two-step method was adopted to sustainably catalyze waste RS into FAL in the CTAB/LAC/Al^{3+}-water system. To improve the FAL yield of the two-step process, the impacts of reaction temperature and reaction duration on FAL yield were deeply explored.

As depicted in Fig. 3a, the pentose content in the system increased from 8.1 g/L to 9.3 g/L by increasing temperature from 140 ℃ to 150 ℃. Upon hydrolysis at 170 ℃, the pentose content exhibited a significant downward trend, and the content of FAL in the system increased. This may be due to the tendency of large amounts of pentose to be converted to FAL at high temperatures, resulting in a decrease in pentose content and accumulation of FAL in the system. In the range of 170 ℃-190 ℃, the pentose content in the system continued to decrease, while the FAL yield remained almost unchanged. In this temperature range, pentose can be quickly transformed into FAL, and the content of FAL cannot be further increased due to the accelerated side-reactions such as polycondensation at high temperatures. As evidenced in Fig. 3b, in 10 min, the pentose yield reached a peak of 9.7 g/L. This may be due to the failure of hemicellulose to be hydrolyzed efficiently in too short reaction time. So, the optimal reaction duration and temperature for preparing pentose were 10 min and 150 ℃, respectively. The pentose in the reaction system was 9.7 g/L. In the FAL synthesis step, the effects of reaction temperature and reaction duration on FAL yield were studied in depth. As revealed in Fig. 3c, the pentose in the system (190 ℃) was rapidly converted into FAL after only 0.5 min. The formed pentose might be rapidly transformed to FAL at elevated temperature. Because the high temperatures not only effectually catalyze the dehydration and cyclization of pentose but also accelerate the side-reactions, a decreased FAL content in the system could be observed when the reaction time was over 5 min. Fig. 3c shows that the FAL yield was 67.3 % (170 ℃, 10 min). Over 20 min, the FAL yield could reach up to 71.6 %. Thus, the appropriate reaction temperature and duration for the FAL synthesis from the prepared pentose were 170 ℃ and 20 min, respectively.

To explore the utilization value of biomass after DES action, the chemical composition of RS was accurately analyzed by chemical composition assay. As revealed in Table S2 (see Supplementary files), there is a big difference between the RS of the one-step process and the two-step process. In the one-step method, RS reacted for 10 min at 180 °C. In the two-step method, RS was filtered and saved after pentose synthesis (150 °C, 10 min), and the pentose solution reacted for 20 min at 170 °C during the furfural synthesis stage. The temperature in the one-step method is higher, leading to more carbohydrate loss. It can be observed that the removal rate of hemicellulose and lignin could reach 100.0 % and 85.5 %, respectively. In contrast, the two-step method yielded 93.7 % cellulose recovery, while most of hemicellulose (nealy 88.0 %) was removed. The two-step method could retain most of the cellulose components, which might be used for further valorization (Bao

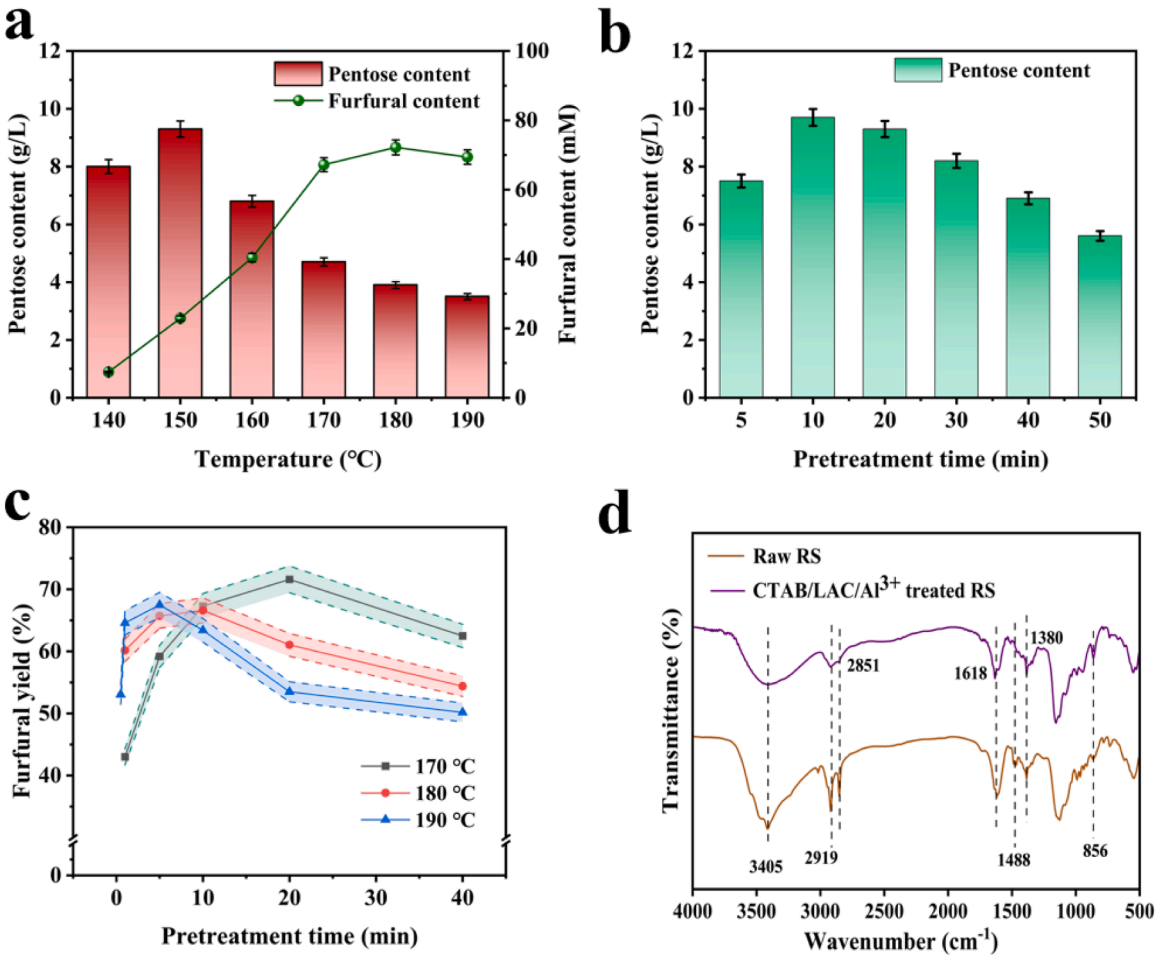


Fig. 3. Effects of temperature on pentose and FAL formation [20 min, CTAB/LAC/$AlCl_3$ 1:4:0.04, DES 50.0 wt%] (a); Effects of reaction duration on pentose formation [170 ℃, CTAB/LAC/$AlCl_3$ 1:4:0.04, DES 50.0 wt%] (b); Effect of temperature and duration on furfural yield [CTAB/LAC/$AlCl_3$ 1:4:0.04, DES 50.0 wt%] (c); The FT-IR image of untreated and treated RS (d).

et al., 2024). So, the two-step synthesis of FAL in the CTAB/LAC/Al^{3+}-water system constructed in this study was considered to have greater advantages, rendering a sustainable and effective strategy for eliminating hemicellulose and lignin and synthesizing FAL.

3.4. Structural characterization of RS

FTIR is commonly adopted to estimate alterations in structural functional groups of biomass (Dharmalingam et al., 2022). As depicted in Fig. 3d, the RS response about 3405 cm^{-1} is tensile vibration caused by hydroxyl groups. A peak near 2919 cm^{-1} is due to the vibration of -CH_2. The peak around 2851 cm^{-1} may be attributed to the -CH_3 group in hemicellulose, and its peak intensity decreases with hemicellulose removal after DES action. The peak of the pretreated RS is considerably weakened at about 1488 cm^{-1}, which may be related to the destruction of the substitution aromatic ring structure of lignin in the reaction (Liu et al., 2023). Under the action of the reaction solvent, the absorption peak almost disappears due to the large amount of lignin removed from the RS (Wu et al., 2023). The absorption peak about 1380 cm^{-1} corresponds to the vibration of -CH in the cellulose structure (Liu et al., 2023). In addition, the absorption peak around 856 cm^{-1} is associated with β-glycosidic bonds in carbohydrates (Chen et al., 2023; Wu et al., 2023). The FT-IR results manifested that the CTAB/LAC/Al^{3+}-water reaction system disintegrated the stubborn lignocellulose structure, dissolved a large amount of lignin and hemicellulose in RS. The surface alterations of lignocellulose can be characterized by SEM (Djafari Petroudy et al., 2021). As evidenced in Fig. 4, the surface structure of the untreated RS is very dense. After DES action, its surface become looser and more pores appear, which is due to the disruption of RS surface with CTAB/LAC/Al^{3+}.

CLSM visually characterize structural alterations and component distribution characteristics of biomass samples through 3D reconstruction and fluorescence analysis (Djafari Petroudy et al., 2021; Hu et al., 2023). By comparing the CLSM microscopy images of the raw RS (Fig. 5a) with the CTAB/LAC/Al^{3+}-treated sample (Fig. 5c) in bright-field, it can be seen that the cell wall structure of the raw material is intact, the cell morphology and the boundaries are clear. After reaction, the cell structures exhibit significant fragmentations, loss of cell wall integrity and a large amount of debris distribution, which may be caused by the severe destruction of the CTAB/LAC/Al^{3+} composite system and the disintegration of lignocellulose structure. At the excitation wavelength of 488 nm, the raw material (Fig. 5b) manifested bright green fluorescence, revealing that its lignin was rich in conjugated aromatic structures. However, the fluorescence intensity of the DES-treated sample declined considerably, and the color was darkened (Fig. 5d). Probably, lignin in RS was eliminated in large quantities during the DES-reaction process, and its carbonyl and conjugated double bonds were destroyed. Meanwhile, the cross-linking network with hemicellulose was dissociated. The above CLSM imaging results intuitively revealed the destructive effect of CTAB/LAC/Al^{3+} action on RS cell wall and lignin elimination from the microscopic scale, rendering strong visual evidence support for the conclusions of preliminary characterization analysis.

BET measurements are often adopted to acquire the surface area and pore characteristics of lignocellulose (Said Al Hoqani et al., 2020). Loose porous structures often render a larger specific surface area, which make cellulose more susceptible to hydrolysis by cellulase attack. In this study, the specific surface area and monolayer adsorption capacity (Q_m) of RS before and after action were compared. As evidenced in Table S3 (see supplementary files), the specific surface area of RS processed by DES sharply increased from 0.897 to 10.97 m^2/g, the monolayer adsorption capacity rose from 0.206 to 2.520 cm^3/g, and the total pore volume increased from 0.0040 to 0.0054 cm^3/g. The micropore volume according to the DFT method did not change, while the average pore diameter increased from 8.07 to 12.23 nm. The correlation coefficients of the BET test of the two samples were 0.996 and 0.987, respectively. Accordingly, the BET was well fitted linearly and the data was highly reliable. The increase in monolayer adsorption capacity indicates that the space available for gas adsorption has increased. The 35 % increase in total pore volume suggests that the overall volume of RS channels has recovered after pretreatment, but the 35 % increase is much smaller than the specific surface area, indicating that obstacles may be covering the inner surface in a thin layer rather than filling the channels. The unchanged micropore volume indicates that the obstacles mainly blocked the mesopores and macropores, likely due to the extremely small pore size of the micropores. The average pore diameter increased by 52 %, indicating that the pretreatment mainly cleared obstacles in the mesopores. The reason for increasing biomass specific surface area and pore size is likely to be due to the separation of hemicellulose and lignin on the surface after reaction, leaving holes of different shapes and sizes and consequently augmenting the specific surface area of biomass and the accessibility of cellulose (Said Al Hoqani et al., 2020). The RS treated by the two-step method of CTAB/LAC/Al^{3+} had a cellulose retention rate of 93.7 %, a lignin removal rate of 66.3 %, and a hemicellulose removal rate of 86.9 %. The effective removal of hemicellulose and lignin confirmed the increase in specific surface area and pore size of the RS.

3.5. Molecular dynamics simulation

Molecular dynamics simulations were adopted to estimate the interactions between the CTAB/LAC/Al^{3+}-water system and the lignocellulose biomass model (Chen et al., 2021; Yin et al., 2022). The electrostatic potential (ESP) of the DES components and the three lignocellulose biomass models was calculated using Multiwfn software. As revealed in Fig. 6, the ESP image renders detailed information about the charge distribution within the lignocellulose and DES components. Specifically, the hydroxyhydrogen atom in lignocellulose behaves as

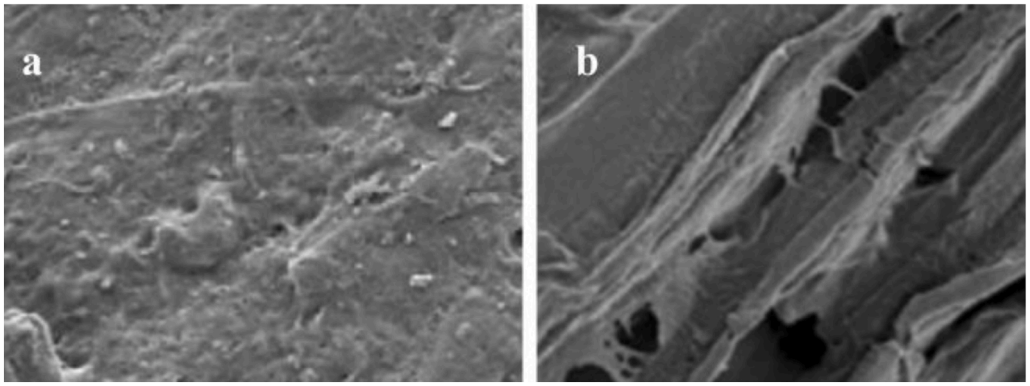


Fig. 4. The SEM image of untreated RS (a) (magnification: 3000 ×) and treated RS (b) (magnification: 2000 ×).

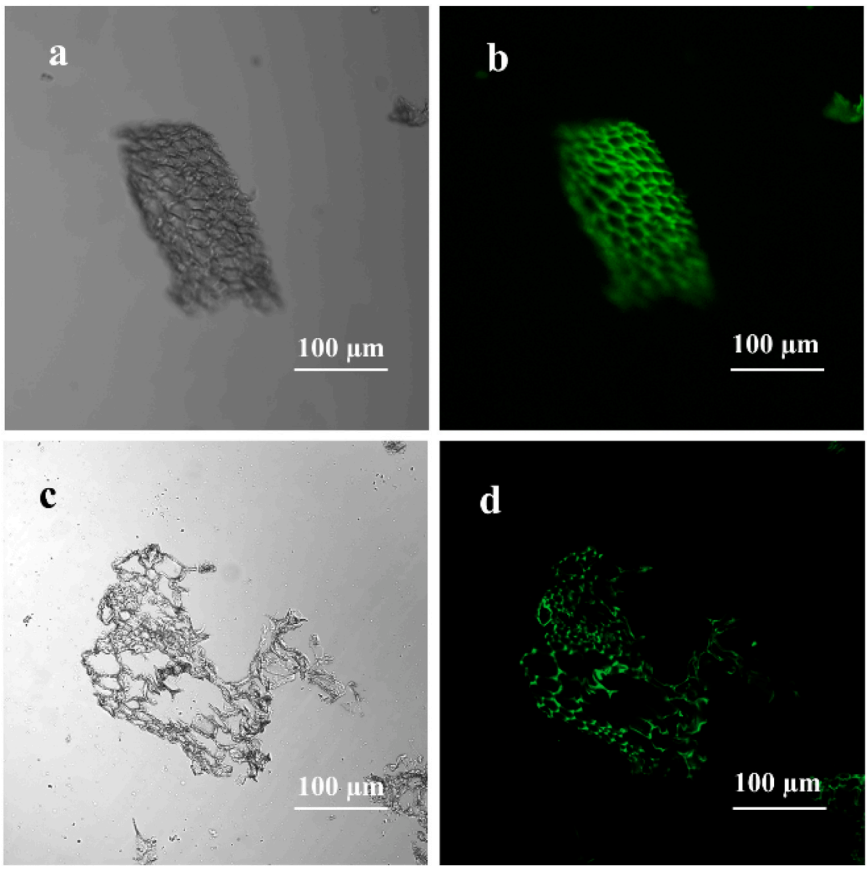


Fig. 5. The CLSM image of untreated and DES-treated RS (a, c: bright field; b, d: 488 nm wavelength excitation).

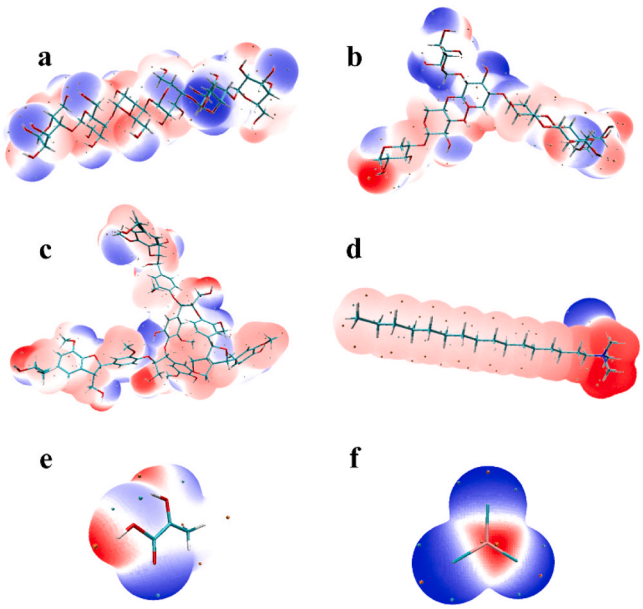


Fig. 6. The electrostatic potential image of cellulose model (a); hemicellulose model (b); and lignin model (c); CTAB (d); LAC (e); and $AlCl_3$ (f).

positive ESP, while the hydroxyhydrogen atom behaves as negative ESP. This charge distribution plays a essential role in the interaction between DES and lignocellulose components. The occurrence of Br^- in CTAB introduces a negative electrostatic potential into the ternary DES system, potentially cleaving hydrogen-bonds in lignin and hemicellulose and severely separating hemicellulose and lignin in lignocellulose. In addition to electrostatic interactions, van der Waals forces also can impact the dissolution and extraction of non-polar compounds in DES system (Wei et al., 2020). In the Van der Waals potential image (Fig. 7), the blue area represents the Van der Waals interactions within the region as attractive. The long alkyl chain of CTAB is encased in a blue region, and this region has a strong van der Waals attraction, which can potentially

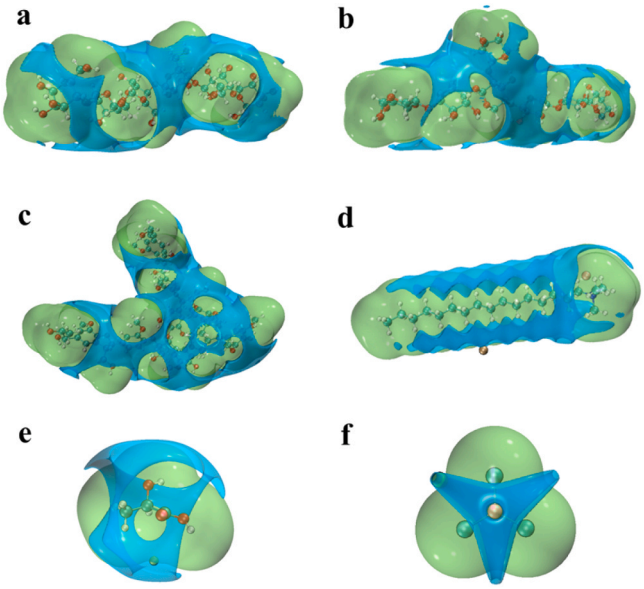


Fig. 7. The van der Waals potential image of cellulose model (a), hemicellulose model (b), and lignin model (c), CTAB (d), LAC (e), and $AlCl_3$ (f).

interact with non-polar substances such as lignin, considerably promoting its elimination. In the lignocellulose models, the lignin model had the strongest dispersion interaction, and the potential energy can reach as low as −0.93, revealing a potentially stronger van der Waals interaction with solvents.

To further apprehend the interactions within DES CTAB/LAC/Al^{3+}-water medium, an independent gradient model (IGM) analysis was performed on the MD simulation system, revealing the weak interactions between models (cellulose, hemicellulose, or lignin model) and CTAB/LAC/Al^{3+} molecules. Hydrogen-bonding is recognized as a relatively potent non-covalent interaction, occurring between hydrogen with strong polar bonds and atoms with lone pairs of electrons (Lefebvre et al., 2017). Carbohydrates (cellulose and hemicelluloses) are both hydroxyl polysaccharides that can serve as both donors and acceptors for hydrogen-bonds.

As manifested in Fig. 8, there is a large dark blue region between the two models and DES, and the density of blue scatter points in the scatter plot is high, displaying that a potent hydrogen-bond interaction between them can occur. The interaction energy between solvent and model can be quantitatively calculated by Gromacs software (Table S4, see Supplementary files). The electrostatic energy (E_{elec}) between cellulose and CTAB/LAC/Al^{3+} is −758.9 kJ/mol, and the electrostatic energy (E_{elec}) between hemicellulose and CTAB/LAC/Al^{3+} is −734.0 kJ/mol, accounting for nearly 83 % and 81 % of the total energy, respectively. As a matter of fact, hydrogen-bond interactions are the main forms of intermolecular electrical interactions, confirming the strong hydrogen-bond interaction between the hemicellulose model and DES. This potent hydrogen-bond interaction may considerably promote the hydrolysis process of hemicellulose in biomass, substantially contribute to the destruction of covalent bonds between sugar molecules, and greatly promote the formation of pentose and FAL (Yaverino-Gutiérrez et al., 2024).

As evidenced in Fig. 8, a large green area between the lignin model and DES is due to a van der Waals interaction. Based on the calculation of intermolecular interaction energy, the Van der Waals interaction (E_{vdW}) between the lignin model and DES is −415.7 kJ/mol, and the Eelec is −605.8 kJ/mol, and the contribution of van der Waals interaction can reach 40.7 %, far exceeding the contribution of the other two models. This is because the structure of lignin consists of rigid aromatic rings linked by aryl ether bonds and flexible side-chains connected by carbon-carbon bonds, lacking strongly polar side-chain groups. Accordingly, it can be seen that van der Waals interactions play an extremely essential role between CTAB/LAC/Al^{3+} and lignin models, interpreting the effectual elimination and dissolution of lignin in biomass in the CTAB/LAC/Al^{3+} system (Wang et al., 2023b). The sustainable production of FAL from lignocellulose has the advantages of wide range of raw materials, low cost and sustainable use of resources. In this research, the newly prepared ternary DES CTAB/LAC/Al^{3+} was adopted to catalyze RS, evidencing that CTAB/LAC/Al^{3+} was a highly efficient reaction solvent for converting hemicellulose in RS into pentose and further into FAL in a short reaction duration.

Utilizing abundant and inexpensive biomass as renewable feedstock

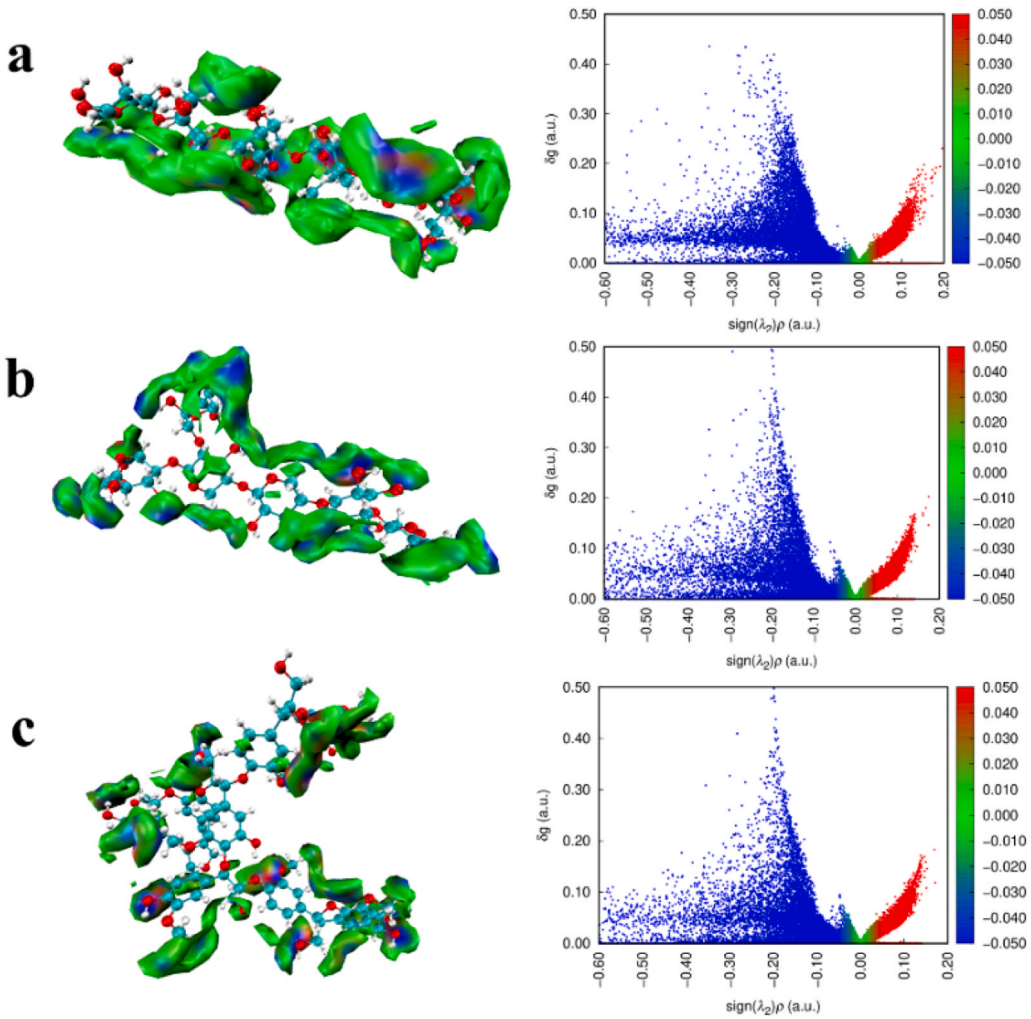


Fig. 8. The IGM isosurface image and scatter plot of cellulose model (a), hemicellulose model (b), and lignin model (c).

to form renewable organic chemicals is crucial for the sustainability of the organic chemical manufacturing industries (Wang et al., 2023b). DESs can be adopted as green reaction medium and catalysts for catalyzing biomass and synthesizing FAL due to their biocompatibility, non-volatility, good thermostability and high catalytic activity (Wang et al., 2023a). Through the two-step method, the pentose solution was obtained from 20 min of RS reaction at 150 ℃, and then the FAL (8.1 g/L) formed through the dehydration of pentose with Al^{3+}-based three-component DES CTAB/LAC/Al^{3+} at 170 ℃ in 20 min, and the FAL yield was as high as 71.6 %. Compared to binary DES without metal ion (CTAC/LAC: FAL 3.8 g/L, 60 min, 160 °C) and Fe^{3+}-based ternary DES (TTAB/LAC/Fe^{3+}: FAL 3.9 g/L, 163 °C, 62 min) (Li and Wu, 2025), this Al^{3+}-based three-component DES CTAB/LAC/Al^{3+} may considerably hydrolyze hemicellulose and remarkably promote the generation of FAL (8.1 g/L) in a short reaction duration (20 min) at 170 ℃. The effectual transformation of RS with CTAB/LAC/Al^{3+} is illustrated in Fig. 9. CTAB/LAC/Al^{3+} is an acidic ternary DES. LAC produces H^+, allowing hemicellulose to be hydrolyzed and produce a chain structure. $AlCl_3$ acts as Lewis acid and catalyzes xylose isomerization to xylose, which is rehydrated to form FAL (Gan et al., 2024). In addition, the Cl^- released by $AlCl_3$ can form hydrogen-bonds with the hydroxyl group in lignin, consequently diminishing the lignin-carbohydrate bonds and allowing a large amount of cellulose and hemicellulose to break through the limit and be converted to pentose (Syakir Bahazin et al., 2025). Molecular dynamics simulations theoretically confirm the strong hydrogen-bond interaction between the RS-hemicellulose model and DES CTAB/LAC/Al^{3+}. In a two-step method, DES can selectively dissolve hemicellulose-derived sugars and generates pentose-rich solutions by disrupting lignin-carbohydrate complexes, followed by dehydration of the xylose-rich solution to yield FAL (Syakir Bahazin et al., 2025). In the traditional one-step catalytic synthesis of FAL, RS is fully mixed and reacted with a CTAB/LAC/Al^{3+}-water system. Hemicellulose undergoes hydrolysis under acidic conditions to produce pentose, which is then dehydrated to synthesize furfural under the action of an acid catalyst. In the one-step method, the two reaction processes of hemicellulose hydrolysis and pentose dehydration cyclization are completed in a single reaction apparatus at once. Although this reduces intermediate steps and simplifies operations, it has downsides such as significant raw material loss, low furfural yield, and serious wastewater pollution (Zhou et al., 2025). Moreover, SEM, CLSM, BET, FT-IR and other methods were adopted to characterize the surface morphology and structure of RS before and after DES action, manifesting that CTAB/LAC/Al^{3+} can strongly destroy the lignin-cellulose-hemicellulose triplet. The surface of the DES-pretreated RS became rougher, the specific surface area increased, and its lignocellulose structure was severely damaged. CTAB in DES, as a surfactant, can also prevent the re-deposition of pseudo-glignin (Tang et al., 2024).

Biorefinery must be economically viable on a large scale to achieve its application (Bao et al., 2024). Among all stages of the biorefining process, the preprocessing stage has the highest variable operating costs, which can be attributed to the high cost of solvents. The recovery and reuse of DES change the raw material costs compared to traditional inorganic acids, significantly impacting the operating costs and making it more cost-effective. The CTAB/LAC/Al^{3+}-water system can be efficiently reused three times (data not shown). Multi-product (value-added) biorefineries are more cost-effective than traditional single product biorefineries. In addition to producing FAL, CTAB/LAC/Al^{3+} should further explore processes that also utilize cellulose and lignin. Furthermore, biorefineries need to install water recycling pipelines, reduce water consumption, and use wastewater treatment systems to promote economic sustainability (Azzouz and Hayyan, 2024).

3.6. Mass flow

A comprehensive mass flow analysis was conducted using 1000 g of RS in this research. As evidenced in **Fig. 10**, RS was pre-treated with CTAB/LAC/Al^{3+}-water under a two-step method. In the first step, RS was treated at 150 °C for 10 min and filtered to obtain solid residues. In the second step, the liquid obtained from the previous step was treated at 170 °C for 20 min. The untreated RS contained 302.4 of glucan, 151.2 g of xylan, and 189.1 g of lignin. The solid recovery rate after CTAB/LAC/Al^{3+} pretreatment was 43.7 %, with a total recovery of 437.3 g of solid residues, which included 282.1 g of glucan, 19.7 g of xylan, and 63.4 g of lignin. In the liquid treated by the two-step method, 108.3 g of FAL was obtained.

4. Conclusions

In this work, RS was catalyzed into FAL in acidic ternary DES-water reaction system, which achieved the good elimination of hemicellulose and lignin in a sustainable way. The prepared ternary DESs acted as solvents and chemocatalysts, where CTAB is the HBA/surfactant, LAC is the HBD/Brønsted acid, and $AlCl_3$ is the Lewis acid catalyst. It was found that the mixture of CTAB/LAC/Al^{3+} (1:4:0.04) and water (50:50, wt/wt) was utilized to transform RS into FAL in 71.6 % yield via a two-step approach. SEM, CLSM, BET and FT-IR of RS before and after DES

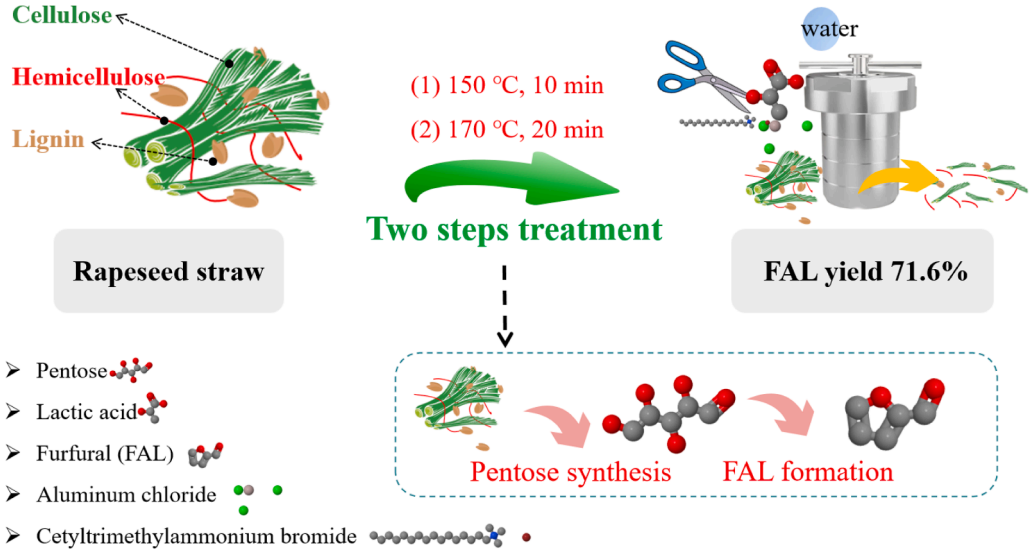


Fig. 9. Reaction of RS in the mixture of CTAB/LAC/Al^{3+} (1:4:0.04) and water (50:50, wt/wt) via a two-step approach.

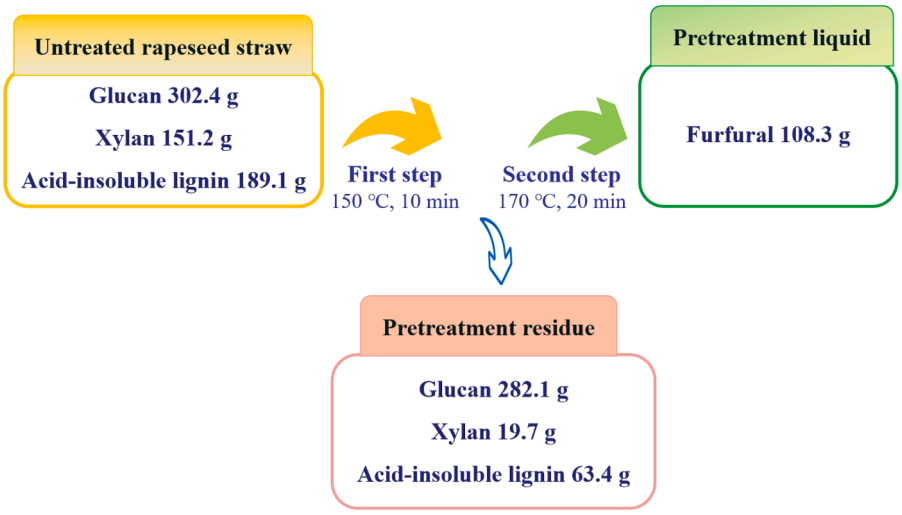


Fig. 10. Mass flow of RS through the pretreatment by CTAB/LAC/Al^{3+}- water.

action manifested that the CTAB/LAC/Al^{3+}-water reaction system destroyed the structure of RS and remarkably promoted the production of FAL. Molecular dynamics simulation revealed the interaction between the RS component model and the CTAB/LAC/Al^{3+} in the DES-water reaction system from the theoretical level. The built CTAB/LAC/Al^{3+}-water rendered a benign reaction system for valorization of RS into FAL.

CRediT authorship contribution statement

Zaiheng Wu: Validation, Software, Resources. **Cuiluan Ma:** Writing – review & editing, Supervision. **Yu-Cai He:** Writing – review & editing, Supervision. **Bo Fan:** Writing – original draft, Software, Methodology, Investigation, Data curation. **Zhengyu Tang:** Visualization, Resources, Methodology, Investigation, Data curation, Conceptualization. **Bingying Yang:** Writing – original draft, Visualization, Resources, Methodology, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors are grateful for the Frontier Technology Research and Development Plan of Jiangsu Province (BF2025080), the National Natural Science Foundation of China (22208031), and the support from the work is kindly supported by the Postgraduate Research & Practice Innovation Program of Jiangsu Province (SJCX25_1684). The authors thank the Analysis and Testing Center (Changzhou University) for analysis of biomass samples.

Informed Consent Statement

Not applicable.

Institutional Review Board Statement

Not applicable.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.indcrop.2025.122042.

Data availability

Data will be made available on request.

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